

EX + C



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: Final Year B.Tech..	Course Name: Elective: Mobile and wireless communication	Course Code: PEC-EC-413
Date: 23.03.2026	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Solve any two questions, assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	What is Indoor Wave Propagation? State main characteristics, factors affecting indoor propagation and applications.	5	CO3	BT2
	b	A wireless transmitter operates at 900 MHz, The distance between transmitter and receiver is 5 km. Calculate the free space path loss.	5	CO3	BT3
Q.2	a	In an indoor wireless system, the path loss at reference distance $d_0 = 1\text{m}$ is 40 dB. The path loss exponent $n=3$ Calculate the path loss at a distance of 10 m.	5	CO3	BT3
	b	Discuss the factors affecting indoor signal propagation in buildings.	5	CO3	BT2
Q.3	a	A transmitter sends 30 dBm power. The indoor propagation path loss is 80 dB. Find the received signal strength.	5	CO3	BT3
	b	A cellular system operates at: Frequency = 1800 MHz, Base station height = 40 m, Mobile height = 1.5 m, Distance = 4 km Find the path loss using COST-231 Hata model.	5	CO3	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 3	CO3 – Explain wave propagation models and characteristics in wireless communication systems
CO 3	CO3 – Apply propagation models to analyze path loss in wireless communication systems
CO 3	CO3 – Apply indoor propagation models to analyze signal attenuation in wireless communication systems
CO 3	CO3 – Explain indoor propagation characteristics and models used in wireless communication systems
CO 5	CO3 – Apply propagation models to evaluate received signal strength in wireless communication systems
CO 6	CO3 – Apply outdoor propagation models to analyze path loss in cellular communication systems



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Department of Electronics and Telecommunication
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: Final Year B.Tech.,

Course Name: Microwave
Engineering

Course Code: PCC-EC-
401

Date: 23.03.2026

Time: 14:30 to 15:15

Max. Marks: 20

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Q.N.			Marks	CO	BT
Q.1	a	A rectangular waveguide has dimensions $a=4$ cm and $b=2$ cm Find the cut-off frequency for dominant TE_{10} mode.	5	CO4	BT3
	b	State Poynting vector theorem and derive an expression for power density and average power density of EM wave.	5	CO2	BT4
Q.2	a	Find the phase velocity of wave and intrinsic impedance of the medium with relative permittivity $\epsilon_r=4$ and $\mu_r=1$	5	CO2	BT3
	b	A rectangular waveguide has dimensions $a=4$ cm, $b=2$ cm Determine the modes that propagate at 10 GHz.	5	CO4	BT4
Q.3	a	A circular waveguide has a radius = 3 cm. Determine the cutoff frequency of the dominant mode (TE_{11}). Given $X_{11}'=1.841$	5	CO4	BT3
	b	Compare Rectangular and Circular Waveguides	5	CO4	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO4	Analyze electromagnetic wave propagation in rectangular and circular waveguides.
CO 2	Analyze electromagnetic wave propagation and power flow using Poynting vector concepts.
CO 2	Analyze electromagnetic wave propagation characteristics in different media.
CO 4	Analyze electromagnetic wave propagation in rectangular and circular waveguides.
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Department of Electronics and Telecommunication
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In-Semester Examination II

Class: Final Year B.Tech..

Course Name: Elective: Mobile and
wireless communication

Course Code: PEC-EC-413

Date: 23.03.2026

Time: 14:30 to 15:15

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Q.3	a	A transmitter sends 30 dBm power. The indoor propagation path loss is 80 dB. Find the received signal strength.	5	CO3	BT3
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CO 3	CO3 – Apply propagation models to analyze path loss in wireless communication systems
CO 3	CO3 – Apply indoor propagation models to analyze signal attenuation in wireless communication systems
CO 3	CO3 – Explain indoor propagation characteristics and models used in wireless communication
CO 5	CO3 – Apply propagation models to evaluate received signal strength in wireless communication systems
CO 6	CO3 – Apply outdoor propagation models to analyze path loss in cellular communication systems



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – II)

Class: B. Tech. Final Year	Subject: Data Mining and Data Warehousing	Subject Code: OEC-EC418
Date: 24/03/2026	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
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Q.N.		Marks	CO	BT
Q.1	Briefly compare the following concepts. You may use an example to explain your point(s). a. Snowflake schema, fact constellation, starlet query model b. Data cleaning, data transformation, refresh c. Enterprise warehouse, data mart, virtual warehouse	8	CO1	BT2
Q.2	Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing and derive general weather patterns in multidimensional space.	7	CO3	BT3
Q.3	What are the differences between the three main types of data warehouse usage: information processing, analytical processing, and data mining? Discuss the motivation behind OLAP mining (OLAM).	5	CO2	BT1

— ALL THE BEST —

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Discuss basic concepts of Data Warehouse fundamentals, data analytics, data mining using professional language associated with data analytics and data mining.
CO 2	Design data warehouse with dimensional modelling and apply OLAP operations.
CO 3	Pre-process the data so that it can be analysed further using sophisticated data analytics and mining algorithms.
CO 4	Identify appropriate data mining algorithms to solve real world problems.
CO 5	Benefit the user experiences towards research and innovation integration.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication
In-Semester Examination II
Academic Year: 2025-26 (Semester-II)

Class: Final Year B.Tech.	Course: Optical Communication Engineering	Course Code: PEC-EC-415
Date: 25.03.2026	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	All Questions are compulsory	Marks	CO	BT
Q.1	Describe the construction and working of Laser Diode. Compare it with LED.	05	CO3	BT2, BT4
Q.2	Explain fiber splicing techniques with advantages and disadvantages.	05	CO3	BT2, BT4
Q.3	Explain the physical principle of operation of a photodiode.	05	CO4	BT2
Q.4	Compare PIN photodiode and Avalanche Photodiode (APD).	05	CO4	BT4

=== ALL THE BEST ===



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Electronics and Telecommunication Engineering

Academic Year: 2025-26 (semester II)

In-Semester Examination II

Class: Final Year B.Tech.

Course Name: Electronic Instrumentation

Course Code: PCC-EC402

Date: 27.03.2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Que. No.	Question	Marks	CO	BT
Q.1	Prove that the capacitive transducer can be used for the measurement of linear displacement.	5	CO2	BT4
Q.2	A compressive force is applied to a structural member. The strain is 5 micro-strain. Two separate strain gauges are attached to the structural member; one is a nickel wire strain gauge having a gauge factor of -12.1 and the other is nichrome wire strain gauge having a gauge factor of 2. Calculate the value of resistance of the gauges after they are strained. The resistance of strain gauges before being strained is 120Ω .	5	CO3	BT3
Q.3	State and explain the laws of thermoelectricity.	5	CO2	BT2
Q.4	Justify the name Linear Variable differential Transformer (LVDT).	5	CO2	BT3

=== ALL THE BEST ===



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE II (Semester-II/ VIII) 2025-26

Final Year B. Tech.

Electronics and Telecommunication Engineering

Internet of Things - IOT

OEC - EC 416

Date: 27-Mar-26

Time: 14:30 to 15:15

Max. Marks : 20

Note: i. Answer **all** questions. ii. Assume suitable data if necessary iii. Use of non-programmable calculators is permitted. iv. This paper contains one page only (back to back).

Q.N.	Question	Marks	COs	BT
Q.1	c. Write a short note on Artchitecture of IoT. d. Differentiate between IOT and Industrial IoT.	10	1, 2	BT 1, 2,
Q. 2	Name the device shown in figure (i). Redraw the figure (i) and label its parts. List its features.	10	2	BT 1, 5, 6

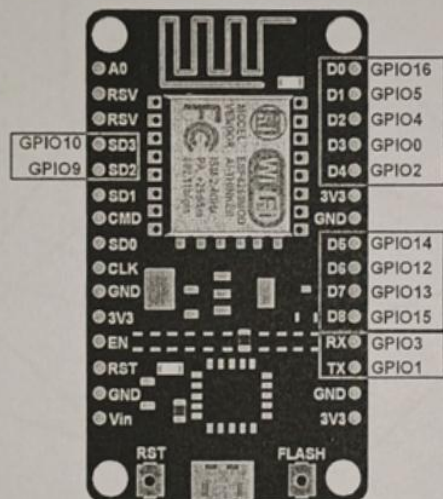


Figure: (i)

*****!!!!!!All the very Best!!!!!!*****

Details below are for Information only



Note: I. Solve ALL Questions

II. Assume suitable data if necessary

Date: 28-March 2026,

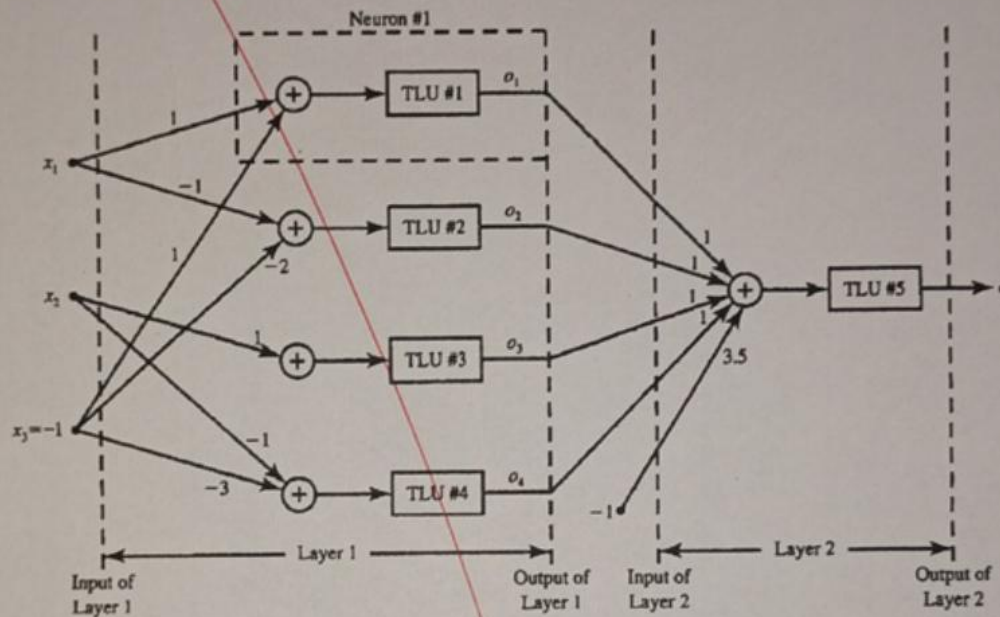
Max marks: 20

Q.1 Construct a neural network to cluster the data into TWO clusters using WTA rule.

Training Data = $\{ [-2, -2]^T, [3, 3]^T, [-2, -3]^T, [2, 3]^T \}$. Assume suitable initial weights for all neurons in your network. Update the weight for ONE iteration. (08)

Q.2 Write winner-take-all (WTA) learning rule and derive the necessary formula to train the network. (06)

Q.3 Find the region on x_1 - x_2 plane where output of the following network is '-1'. (06)



Course Outcomes (COs):

1. Understanding the basics of Artificial Intelligence, artificial neural network, their limitations, basic pattern analysis tasks such as classification and clustering, learning and adaptation using the learning rules, implementation of learning rule.
2. Describe the concepts of feed forward neural networks using single layer and multilayer networks to solve classification problem, and its implementation, single layer feedback networks to study the concept of memory using neural networks.
3. Analyze and implement the applications of neural networks in character recognition and control systems.
4. Understand Auto associative neural networks, Pattern storage and retrieval, Hopfield model, recurrent neural networks
5. Analyze Bayesian neural networks, Radial basis function networks
6. Understand self-organizing maps and recent trends in neural networks

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BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6- Create

Q. No	CO No	BT No
1	1	3
2	1	2
3	2	5